

AK High-Dimensional Projection Structural Theory

Appendix TB

Toy Bridge, Counterexample, and Regression Portfolio

v21 global re-audit of the frozen TB-v20 theorem pack

with repaired CM bindings, dual-Core guard-rails, strict descent validation,
target-relative Return records, and derived toy closure

Theory version	v21.0.0
Appendix version	TB-v21.0.0
Document role	Toy validation, counterexample, regression, and demonstration appendix
Evidence status	Mixed proved / conditional / operational / historical predecessor
Canonical theorem owners	Main, Foundation A–G, DG, CM, HT, and UB as declared
Claim Register	pending final CR-v21
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Current authority. Appendix TB-v21 owns only explicitly bounded toy-category theorems, counterexamples, regression witnesses, source-specific implementations, and validated demonstrations. Main Chapters 2–9 own the generic Core, verification, Bridge, and Return calculi; Appendix DG owns generic finite-diagram and descent theorems; Appendix CM owns reusable Core corollaries and implementation lemmas; Appendix UB owns the generic component and validated-return portfolio. Repetition in TB never creates a competing canonical theorem.

Frozen-predecessor rule. The complete fifty-three-item TB-v20 corpus is retained below as an immutable historical source. Its nine dangling CM-v20 references and its physical-window mesh formula are not current v21 evidence. The v21 synchronization layer supplies the only permitted repaired readings.

Nonpromotion rule. A toy pass proves no arithmetic, PDE, geometric-equivalence, cryptographic, or other external-domain theorem. Readiness, effectivity, target-relative Return strength, workflow completion, artifact closure, migration, and Claim registration remain distinct sorts.

Abstract

Appendix TB-v21 is the source-specific toy validation and regression layer of AK-HPDST v21.0.0. It preserves the complete fifty-three-item TB-v20 corpus while rebasing current use onto the dual-Core architecture, the repaired CM-v21 consumer contract, Appendix DG strict descent, Chapter 8 proof-DAG semantics, and Chapter 9 target-relative Bridge/Return calculus. Nine nonexistent CM-v20 dependencies are quarantined and re-bound without fabricating historical UIDs. The defective physical-window mesh statement is replaced by an exact effective-domain theorem.

Twelve v21-local results are added: strict-transition budgets for finite-dimensional and finite-length towers; finite direct-sum closure of exact toy towers; finite-chain Type IV accumulation; terminal-clean two-out-of-three and short-five principles; corrected effective-domain detector meshes; instance-relative finite witness extraction; strict finite-diagram exact globalization; compatibility of strict descent with eventual constancy; independent target-product toy Returns; and a sharp objectwise-isomorphism/naturality counterexample. These are toy implementations or consequences, not new generic Main/DG/UB theorem authorities. The global re-audit further aligns the product Return with the final UB product-category theorem, expands the lossless implementation schema, fixes targetwise partial-Return semantics, and makes descent, tail, and Type IV prerequisites explicit.

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Part I. v21 constitutional synchronization, repaired bindings, and derived toy theorems

1 v21 scope, ownership, and typed consumer contract

v21 Definition 1 (TB-v21 role and ownership). *Appendix TB-v21 has the primary role*

toy validation, counterexample, regression, and demonstration portfolio.

Its admissible local outputs are toy source theorems, source-specific bridge implementations, exact counterexamples, finite regression witnesses, validated demonstration records, and consumer test cases. Generic statements remain owned by their Main/Foundation/DG/CM/UB authorities.

v21 Policy 2 (TB-v21-OWN-01: no third theorem authority). *A TB item may cite, instantiate, or regress a canonical theorem, but it does not republish that theorem under a competing identity. In particular, finite diagrams and descent are owned by Main Chapter 7 and Appendix DG; micro-theorem calculus by Appendix CM; generic Bridge/Return composition by Main Chapter 9 and Appendix UB; and verifier handoff by Main Chapter 8.*

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-OWN-01
Role / status	operational_policy / operational
Invocation / conformance	operational_rule / conformant
Source bindings	Main Chapters 7–9; Appendices DG, CM, UB
Source binding	v21 direct owner binding; final CR pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared
Failure routes	FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry	new v21 ownership constitution.

v21 Definition 3 (Lossless toy implementation record). *A current TB implementation is serialized as*

$$\mathfrak{T}_{\text{TB}} = (\text{id}, \text{owner}, \mathcal{C}_{\text{src}}, \mathcal{C}_{\text{tgt}}, F, \text{var}, \text{coeff/endpt}, \text{Hyp}, \text{mode}, \text{stage}, \text{Out}, \Gamma, \mathfrak{S}, \text{Fail}, \text{Test}, \mathcal{A}).$$

Here var records variance, coeff/endpt records coefficient and endpoint conventions, Γ is the propagated assumption context, $\mathfrak{S}$ is a downward-closed target-relative strength set, Fail records one primary failure owner and its structured failure route, Test is a regression witness, and $\mathcal{A}$ is the immutable artifact bundle. Projection to a shorter record is permitted only when lossless reconstruction of every omitted field is proved.

v21 Policy 4 (TB-v21-TYPE-01: status and result-sort separation). *Atomic verifier status is one of*

pass, reject, undefined, not_invoked.

A P2DG input has its own four-valued readiness axis

ready, incomplete, inconsistent, not_invoked.

The later handoff tokens descent_input_ready and bridge_input_ready belong to different fields.

The specialized Type IV diagnostic status is separately

not_invoked, undefined, certified_zero, obstructed.

For a fixed target, the Bridge-result sort is

reject, undefined, obstructed, partial_return, regularized_return, return.

No lexical overlap identifies these sorts, and no readiness or diagnostic token is itself a mathematical Return. In particular, the frozen TB-v20-052 sentence about a missing invoked arrow is read targetwise: the exact declared target is **undefined**; a separately declared weaker target may still receive **partial_return** only through its own complete route.

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-TYPE-01
Role / status	operational_policy / operational
Invocation / conformance	operational_rule / conformant
Source bindings	Main Chapters 2, 8, 9; CM-v21 interface contract
Source binding	v21 direct typed binding; final CR pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared
Failure routes	FR-MISSING, FR-NONVERDICT, FR-RETURN-MISSING, FR-T5
Repair ancestry	new v21 sort-separation policy.

v21 Policy 5 (TB-v21-CM-01: repaired CM consumer bindings). The frozen TB-v20 strings CM-v20-075, 077, 079, 080, 081, 082, 083, 085, and 086 do not resolve and are never theorem evidence. Current v21 consumption uses the source-correct bindings in Table 1. This repair does not mutate the frozen TB-v20 artifact.

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-CM-01
Role / status	source_binding_repair / operational
Invocation / conformance	consumer_contract / conformant
Source bindings	CM-v20-001-073; CM-v21-DER-05, DER-11, DER-12, GOV-01
Source binding	v21 repair matrix; frozen predecessor unchanged
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry	repairs nine dangling historical source tokens.

Table 1: Current TB-v21 repair of dangling TB-v20-to-CM bindings

Frozen token	Current source-correct edge	Permitted v21 reading
CM-v20-075	CM-v20-051, 052; CM-v21-DER-05	uniform-excess completeness and the effective-domain minimal mesh

Frozen token	Current source-correct edge	Permitted v21 reading
CM-v20-077	CM-v20-057, 059, 060	synthetic duality only with separate natural zero-reflection and nonpromotion
CM-v20-079	CM-v20-063, 067	terminal-zero/transient separation
CM-v20-080	CM-v21-DER-11	endpoint-object data do not determine morphism defects
CM-v20-081	CM-v20-064	stage-typed transient-defect detection
CM-v20-082	CM-v21-DER-12	local-zero/global-nonzero partition witness
CM-v20-083	CM-v20-068	retained-bar coverage for atlas globalization
CM-v20-085	CM-v20-059, 062	natural zero-reflection and detector-to-higher-edge discharge
CM-v20-086	CM-v20-073; CM-v21-GOV-01	manifest and source-binding closure; registration remains nonproof

v21 Policy 6 (TB-v21-DG-01: one active descent lane and no local-to-global shortcut). *A toy finite-diagram or atlas demonstration invokes exactly one active descent lane among*

`strict_alexandrov_sheaf`, `strict_bounded_complex`, `model_certified_derived`.

Every invoked descent route supplies the canonical nine-field `DescentInput` record, including object data, transition maps, cocycles, homotopies when applicable, window and comparison data, coherence model, effectivity theorem, and artifacts. Core-P detector evidence is not relabeled as a diagram probe or descent-coherence certificate. Local zero, local isomorphism, or local eventual constancy globalizes only through an explicit coverage/effectivity theorem with the actual finite diagram, cocycle, compatibility data, and artifacts; no cross-lane borrowing is permitted.

TB UID	<code>ak-hpdst:21.0.0:TB-v21:TB-v21-DG-01</code>
Role / status	<code>operational_policy</code> / <code>operational</code>
Invocation / conformance	<code>operational_rule</code> / <code>conformant</code>
Source bindings	Main Chapter 7; Appendix DG; CM-v20-068, 069
Source binding	v21 direct DG consumer binding
Comparison policy	<code>theorem_controlled_nonmetric</code>
Allowed runtime modes	<code>theorem_controlled_nonmetric</code> , <code>not_compared</code>
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-NONVERDICT, FR-T5
Repair ancestry	new v21 descent firewall.

v21 Policy 7 (TB-v21-VAL-01: validated demonstration and no benchmark backflow). *A TB demonstration may enter Chapter 10 as a validated toy return, counterexample, or regression witness only after its complete assumptions, target, strength, failure route, and replay artifact are fixed. Success on a toy benchmark does not strengthen the generic theorem, enlarge an external target class, or feed assumptions backward into Main, DG, CM, HT, or UB.*

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-VAL-01
Role / status	validation_policy / operational
Invocation / conformance	operational_rule / conformant
Source bindings	Main Chapter 10; Main Chapter 9; Appendix UB
Source binding	v21 direct validation binding
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared
Failure routes	FR-MISSING, FR-NONVERDICT, FR-RETURN-MISSING, FR-T5
Repair ancestry	new v21 benchmark-quarantine policy.

2 v21-local derived toy theorems

v21 Theorem 8 (TB-v21-DER-01: strict-transition budget for injective finite-dimensional towers).

Let $V_0 \xrightarrow{f_0} V_1 \rightarrow \dots$ be a tower of finite-dimensional vector spaces with every f_n injective and $\dim V_n \leq D$. Then

$$\#\{n \mid f_n \text{ is not an isomorphism}\} \leq D - \dim V_0.$$

Consequently the tower is eventually a tower of isomorphisms. The dimension bound controls the number of strict transitions, not the numerical index of the last strict transition.

Proof. An injective non-isomorphism strictly raises dimension by at least one. The total possible increase is at most $D - \dim V_0$. Strict transitions may be separated by arbitrarily long strings of isomorphisms, so no stage-index bound follows from D alone. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-01
Role / status	toy_source_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	TB-v20-007; finite-dimensional rank calculus
Source binding	v21 local derived theorem
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T5
Repair ancestry	strengthens the qualitative stabilization witness without changing its scope.

v21 Theorem 9 (TB-v21-DER-02: strict-transition budget for finite-length module towers). Let $M_0 \hookrightarrow M_1 \hookrightarrow \dots$ be a tower of finite-length modules with $\text{len}_R(M_n) \leq L$. Then

$$\#\{n \mid M_n \rightarrow M_{n+1} \text{ is not an isomorphism}\} \leq L - \text{len}_R(M_0).$$

The analogous epimorphism tower has at most $\text{len}_R(M_0)$ strict quotient transitions.

Proof. Each strict monomorphism increases composition length by at least one. Each strict epimorphism decreases it by at least one. Additivity of length gives the bounds. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-02
Role / status	toy_source_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	TB-v20-009; Jordan–Hölder length calculus
Source binding	v21 local derived theorem
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T5
Repair ancestry	quantitative strict-transition refinement.

v21 Theorem 10 (TB-v21-DER-03: finite direct-sum closure of exact toy towers). *For $1 \leq j \leq m$, let $Q_{\bullet}^{(j)}$ be an exact eventually constant toy tower with stabilization stage N_j , declared apex $Q_{\infty}^{(j)}$, and isomorphic colimit-to-apex artifact. In an additive category with the declared colimits, equip the finite direct-sum tower*

$$\bigoplus_{j=1}^m Q_0^{(j)} \longrightarrow \bigoplus_{j=1}^m Q_1^{(j)} \longrightarrow \dots$$

with apex $\bigoplus_j Q_{\infty}^{(j)}$ and the coordinatewise induced colimit-to-apex artifact. It stabilizes after $N = \max_j N_j$, and that artifact is an isomorphism. On a common terminal-tame scope its aggregated Type IV diagnostics vanish.

Proof. After N , every component transition is an isomorphism, hence so is the finite direct sum. Finite biproducts preserve isomorphisms, and the declared colimit of an eventually constant diagram is its stable value. Kernels and cokernels of the direct-sum apex map vanish; equivalently, use finite additivity of the diagnostics. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-03
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	TB-v20-004, 014; CM-v20-035, 070
Source binding	v21 local derived theorem
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-T5
Repair ancestry	finite closure of exact toy tower artifacts.

v21 Theorem 11 (TB-v21-DER-04: finite-chain accumulation of terminal-generic defects). *Let*

$$M_0 \xrightarrow{f_1} M_1 \xrightarrow{f_2} \dots \xrightarrow{f_m} M_m$$

be terminal-tame actual morphisms on one common terminal scope. Then

$$\mu(f_m \cdots f_1) \leq \sum_{j=1}^m \mu(f_j), \quad \nu(f_m \cdots f_1) \leq \sum_{j=1}^m \nu(f_j).$$

Thus a finite chain of terminal-clean toy comparisons is terminal-clean.

Proof. Iterate the two-map composition inequalities. The common terminal scope is required so that all dimensions are evaluated in the same stabilized regime. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-04
Role / status	toy_bridge_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-036; CM-v21-DER-09; TB-v20-015
Source binding	v21 local derived theorem
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	finite-chain implementation of the canonical defect calculus.

v21 Theorem 12 (TB-v21-DER-05: terminal-clean two-out-of-three). *Let $M \xrightarrow{f} N \xrightarrow{g} P$ be actual morphisms. Assume M, N, P and the kernel and cokernel profiles of f, g , and gf are terminal-tame on one common stabilized tail. If any two of f, g , and gf are terminal-clean, then the third is terminal-clean.*

Proof. Terminal cleanliness is equivalent to the induced stabilized-tail map being an isomorphism. Isomorphisms satisfy two-out-of-three. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-05
Role / status	toy_bridge_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v21-DER-07; TB-v20-013, 015
Source binding	v21 local derived theorem
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	toy implementation of terminal-tail two-out-of-three.

v21 Theorem 13 (TB-v21-DER-06: terminal short-five principle). *In a commutative diagram with short exact rows and a common terminal-tame scope, if any two of the three vertical morphisms are terminal-clean, then the third is terminal-clean.*

Proof. Pass to the common stabilized tail. If the outer maps are isomorphisms, apply the short five lemma. If the middle and right maps are isomorphisms, identify the left terms as kernels; if the left and middle maps are isomorphisms, identify the right terms as cokernels. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-06
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-037; CM-v21-DER-08; TB-v20-016
Source binding	v21 local derived theorem
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-T5
Repair ancestry	toy short-exact implementation.

v21 Theorem 14 (TB-v21-DER-07: corrected effective-domain family mesh). *Let $W = [a, b)$, $\tau > 0$, $c = b - \tau$, and $L_{\text{eff}} = \max\{c - a, 0\}$. Suppose every retained window-relative bar in a declared family has excess at least $\gamma > 0$ beyond τ . Choose $0 < h < \gamma$. If $L_{\text{eff}} > 0$, set*

$$N = \left\lceil \frac{L_{\text{eff}}}{h} \right\rceil, \quad T_h = \left\{ a + \frac{jL_{\text{eff}}}{N} : 1 \leq j \leq N - 1 \right\}.$$

The interior stencil has augmented mesh, computed using the boundary markers a, c , at most h , and uses exactly

$$\max \left\{ 0, \left\lceil \frac{L_{\text{eff}}}{h} \right\rceil - 1 \right\}$$

actual query points. It is complete for the family, and every refinement remains complete. If $L_{\text{eff}} = 0$, every window-clipped finite bar has length at most τ ; the repaired finite-bar profile is identically zero and the separate empty-effective-domain lane requires no query. Neither boundary marker is silently counted as a detector query.

Proof. The interior points divide $[a, c]$ into N equal augmented gaps of length $L_{\text{eff}}/N \leq h$. Every retained bar has an active-start interval of length at least $\gamma > h$, hence contains a stencil start. Minimality of the query count follows because m interior points create only $m + 1$ augmented gaps whose lengths sum to L_{eff} . $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-07
Role / status	conditional_toy_detector_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-051, 052; CM-v21-DER-05
Source binding	v21 replacement for the frozen TB-v20-035 current reading
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	corrects physical-window counting to the effective detector domain.

v21 Theorem 15 (TB-v21-DER-08: instance-relative finite retained-bar witness extraction). *Let B be one fixed finite repaired barcode with r retained bars. Suppose a family of local detector scopes covers every retained bar and supplies at least one complete witness for each bar. Then there exists a subfamily of at most r local witnesses such that simultaneous zero of those selected detector values implies $B = 0$.*

Proof. Choose one local witness for each retained bar and remove duplicates. If all selected values were zero while a retained bar remained, its chosen complete witness would be nonzero. The bound is instance-relative and does not assert a family-uniform MinCert value. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-08
Role / status	toy_certificate_corollary / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-050, 068; TB-v20-048
Source binding	v21 local derived theorem
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry	instance-level certificate extraction only.

v21 Theorem 16 (TB-v21-DER-09: strict finite-diagram exact globalization). *Let I be a finite category, let $F, G : I \rightarrow \mathcal{A}$ be actual finite diagrams, and let $\eta : F \Rightarrow G$ be a natural transformation whose every component is an isomorphism. Then η is a natural isomorphism. Suppose one active strict descent lane supplies an effectivity construction that functorially descends F , G , η , and componentwise inverse data to an actual descended morphism. The descended comparison is then an isomorphism. When Type IV is invoked and the descended source, target, kernel, and cokernel records are terminal-tame on the same scope, its valid diagnostics are zero.*

Proof. The componentwise inverses satisfy naturality by conjugating the naturality squares of η . Hence η is an isomorphism in the finite diagram category. An effective descent construction preserves isomorphisms; the descended kernel and cokernel vanish. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-09
Role / status	conditional_toy_DG_validation / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	Main Chapter 7; Appendix DG; TB-v21-DG-01
Source binding	v21 toy validation of strict finite-diagram globalization
Comparison policy	theorem_controlled_nonmetric
Allowed runtime modes	theorem_controlled_nonmetric
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-T5
Repair ancestry	does not republish the generic DG effectivity theorem.

v21 Theorem 17 (TB-v21-DER-10: strict descent commutes with eventual-constancy validation). *Fix one active strict descent lane over a finite index category. For each n , let compatible local data $E_{\alpha,n}$ descend effectively to E_n , and let all transition maps be morphisms of descent data. If there is N such that every local transition $E_{\alpha,n} \rightarrow E_{\alpha,n+1}$ is an isomorphism for $n \geq N$, then the descended global tower is eventually constant. Assume its declared colimit exists. If compatible local apex agreements descend to an isomorphism $E_N \xrightarrow{\sim} E_\infty$ and the global colimit-to-apex artifact is the induced comparison, then that comparison is an isomorphism.*

Proof. For $n \geq N$, the inverse local maps remain compatible with the strict descent data, so the descended transition is an isomorphism. The global tower is therefore eventually constant.

Compatible local apex isomorphisms descend to a global apex isomorphism, and the exact eventual-constancy bridge applies. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-10
Role / status	conditional_toy_DG_bridge / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	Appendix DG strict lane; CM-v20-070; TB-v20-004
Source binding	v21 local cross-component validation
Comparison policy	theorem_controlled_nonmetric
Allowed runtime modes	theorem_controlled_nonmetric, exact
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-RETURN-MISSING, FR-T5
Repair ancestry	strict-lane theorem only; no cross-lane borrowing.

v21 Theorem 18 (TB-v21-DER-11: independent target-product toy Return). *Suppose two target routes share one certified upstream toy sub-DAG, have disjoint terminal target nodes, and return independently to target records $R_1 \in \mathcal{C}_1$ and $R_2 \in \mathcal{C}_2$ at coordinate strengths σ_1 and σ_2 . Assume their normalized union is acyclic, every shared node has one identity, status, and failure owner, and no cross-target compatibility obligation is requested. Then the product-record target*

$$(R_1, R_2) \in \mathcal{C}_1 \times \mathcal{C}_2$$

returns at product strength (σ_1, σ_2) . If the component supported lower sets are S_1 and S_2 , the product supported set is $S_1 \times S_2$, not an untyped intersection. Conversely, the registered projections recover the component Returns. No equation, coherence cell, diagonal condition, or common-target product object follows without an additional theorem.

Proof. Pair the two returned target records and their artifacts on the common certified upstream source. Coordinatewise target relations make the strength coordinates independent, so the supported lower set is the Cartesian product. Shared upstream nodes are identified rather than duplicated, preserving single-valued failure ownership. The registered projections recover the original Returns. Any cross-target relation is a separate component and is not discharged here. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-11
Role / status	conditional_toy_return_validation / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	Main Chapter 9; Appendix UB, UB-v21-DER-05
Source binding	v21 toy validation; generic owner unchanged
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared
Failure routes	FR-MISSING, FR-RETURN-MISSING, FR-T5
Repair ancestry	target-relative product record only.

v21 Theorem 19 (TB-v21-DER-12: objectwise isomorphism does not manufacture naturality). *Let $[1]$ be the category with one nonidentity arrow $0 \rightarrow 1$. Define functors $F, G : [1] \rightarrow \mathbf{Vect}_k$ by*

$$F(0) = F(1) = G(0) = G(1) = k, \quad F(0 \rightarrow 1) = \text{id}_k, \quad G(0 \rightarrow 1) = 0.$$

The object values are pairwise isomorphic, but there is no natural isomorphism $F \Rightarrow G$. Hence objectwise toy equivalence, equal dimension data, or endpoint-object distance zero does not establish a diagrammatic bridge.

Proof. If $\alpha : F \Rightarrow G$ were a natural isomorphism, naturality would give $\alpha_1 \circ \text{id}_k = 0 \circ \alpha_0 = 0$, so $\alpha_1 = 0$, contradicting invertibility. $\square$

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-DER-12
Role / status	toy_counterexample / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	Main Chapter 7 naturality boundary; CM-v21-DER-11
Source binding	v21 local regression witness
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	new objectwise-versus-natural counterexample.

3 Migration, TB consumer disposition, and closure

v21 Definition 20 (Current disposition of the frozen TB-v20 corpus). *The fifty-three predecessor UUIDs remain immutable source locators. Their v21 disposition is one of:*

1. *inherited-validation:* statement remains a valid toy theorem or counterexample at unchanged scope;
2. *owner-relocated:* generic theorem authority belongs to Main, DG, CM, or UB while TB remains a validation locator;
3. *binding-repaired:* mathematical reading is retained but current source edges are replaced by Table 1;
4. *statement-replaced:* the predecessor is regression evidence only and a v21 theorem supplies the current mathematical statement.

The only statement-replaced item is TB-v20-035, whose current replacement is TB-v21-DER-07. No other predecessor theorem is strengthened silently.

v21 Policy 21 (TB-v21-GOV-01: lossless register and replay closure). *A current TB result is reusable only when its local ID, owner, source/target schema, realization and variance, coefficient/endpoint conventions, hypotheses, comparison mode, detector stage, propagated assumptions, target-relative strength, primary failure owner, artifacts, and regression witness serialize losslessly. Every infinite quantifier selects exactly one declared tail mechanism for that proof-DAG run; eventual constancy, finite-state reduction, Noetherian reduction, and summability are not blended without a separate theorem. Operational replay, semantic replay, and mathematical proof are separate levels. Claim registration is a later governance action and creates no theorem.*

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-GOV-01
Role / status	serialization_policy / operational
Invocation / conformance	operational_rule / conformant
Source bindings	Main Chapter 8; CM-v20-073; CM-v21-GOV-01; Appendix UB
Source binding	v21 local governance record
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared
Failure routes	FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry	new v21 serialization and replay rule.

v21 Theorem 22 (TB-v21-CLS-01: v21 toy portfolio closure). *Assume all invoked Main/Foundation/DG/CM/UB premises, current source bindings, target records, and artifacts are valid. Then Appendix TB-v21 supplies a conservative portfolio consisting of the frozen fifty-three-item predecessor corpus, the repaired current-reading map, twelve v21-local derived toy theorems, and explicit nonpromotion boundaries. No item changes a generic theorem owner, constructs an unproved external interface, or promotes a toy result beyond its declared target and strength.*

Proof. The ownership and type policies fix the permitted roles. Each derived theorem is proved above in its declared toy regime. The repair matrix replaces only invalid current dependency edges, while the frozen corpus remains unchanged. The globally reaudited product-Return theorem uses product strength and Cartesian lower sets, and the descent theorems require their explicit effectivity and artifact hypotheses. Target and strength ceilings are explicit, so no stronger conclusion follows. □

TB UID	ak-hpdst:21.0.0:TB-v21:TB-v21-CLS-01
Role / status	conditional_package_closure / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	TB-v21-OWN-01, TYPE-01, CM-01, DG-01, VAL-01, DER-01–12, GOV-01
Source binding	v21 local closure; final CR pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared
Failure routes	FR-MISSING, FR-NONVERDICT, FR-RETURN-MISSING, FR-T5
Repair ancestry	new v21 closure theorem.

Frozen TB-v20 UIDs	53 / 53 preserved
Dangling historical CM tokens	9 quarantined and source-correctly repaired for v21
Statement replacement	TB-v20-035 only; replacement TB-v21-DER-07
v21 local identities	19
v21 derived toy theorems	12
Generic theorem ownership	unchanged
Claim Register	pending final CR-v21

Part II. Complete frozen TB-v20 predecessor corpus under the v21 current-reading overlay

Guard-rail. The following corpus is reproduced as immutable predecessor evidence. Its nine dangling CM-v20 tokens are historical defects, not current source bindings. TB-v20-035 is retained only as regression evidence; TB-v21-DER-07 is the current mesh theorem. All other current readings are constrained by Part I.

4 Scope, Dependency Baseline, and Theorem Boundary

This appendix is a conservative v20 reconstruction of the canonical TB-v19 predecessor. The inherited exact-tower, finite-length stabilization, degree-one eligible edge, terminal-generic defect, interleaving, metric-gap, and audit theorems are preserved without weakening their hypotheses. The v20 extension adds finite detector examples and no-go results, synthetic and eligible higher-Ext toy bridges, explicit transient-defect counterexamples, detector-safe atlas and Restart boundaries, and exact finite-to-infinite toy reductions.

Principle 4.1 (Toy bridge sovereignty). *A toy bridge theorem may transport a proved source-side toy condition to a Core-visible conclusion only after the source category, realization route, monitored scope, detector stage when invoked, comparison artifact, completeness source, and required evidence have been declared. The conclusion is limited to the toy semantics actually proved.*

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-001
Role / status	operational_policy / operational
Invocation / conformance	operational_rule / conformant
Source bindings	CR-G-001, CR-G-002, CR-G-006, CR-G-007, CR-G-017, CR-G-018
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry	TB-v19-001; conservatively inherited and re-bound to v20 sources.

Principle 4.2 (No external-domain or naturality upgrade). *No theorem in Appendix TB-v20 proves BSD, ABC, RH, an Iwasawa main conjecture, Navier–Stokes regularity, a geometric or categorical equivalence, a cryptographic security theorem, or an AK-to-external Return Theorem. A synthetic or deliberately engineered toy functor proves only a theorem in that toy category and cannot substitute for a purpose-faithful external realization.*

Definition 4.3 (Immutable dependency baseline). *The normative dependency order is:*

1. the immutable repaired v19 lower baseline, used only as repair ancestry;
2. AK-HPDST v20.0.0 Main Chapters 1–8;
3. v20 Foundation Appendices A–G;
4. the CR-v20 baseline register for claim-use governance;
5. the final CM-v20 artifact as the direct source of reusable micro-theorems;

6. *this TB-v20 artifact and its co-normative JSON/CSV local register.*

The CR baseline intentionally leaves CM-v20 and TB-v20 unpromoted. This appendix therefore binds directly to their immutable artifacts and records formal CR promotion as a later governance receipt rather than fabricating completed promotion.

Remark 4.4 (Canonical source identities). *The package manifest binds the exact source artifacts by full SHA-256. The body does not use basenames, paths, shortened digests, or upload timestamps as mathematical authority.*

Guard-rail. Registration is not proof. A complete manifest is not proof. A successful XeLaTeX build is not proof. Each theorem below is valid only at the strength of its displayed hypotheses and source bindings.

5 Standing Repaired Core Regime

Fix a field k . Work in the constructible one-parameter persistence regime

$$\text{Pers}_k^{\text{cons}} \subset [(\mathbb{R}, \leq), \text{Vect}_k]$$

and with finite-type filtered objects in $\text{FiltCh}(k)$ whenever a filtered realization is used.

For a monitored degree i , right-open window W , and threshold $\tau > 0$, set

$$B_{i,W}(F) := W_W P_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

Windowing precedes hard deletion. Threshold clearance is measured on $B_{i,W}(F)$ before deletion.

Definition 5.1 (Canonical comparison modes and detector stages). *Every consumed comparison resolves to exactly one runtime mode:*

exact, metric_gap_certified, not_compared.

The mode not_compared is permitted only when no comparison inference is consumed. Every detector additionally declares exactly one stage among

windowed_prethreshold, repaired_postthreshold, kernel_defect, cokernel_defect.

Neither comparison mode nor detector stage may be silently changed by a toy realization.

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-002
Role / status	operational_policy / operational
Invocation / conformance	operational_rule / conformant
Source bindings	CR-G-004, CR-G-010, CR-G-014, CM-v20-015, CM-v20-041
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-002; conservatively inherited and re-bound to v20 sources.

Definition 5.2 (Typed Type IV scope). *For each invoked monitored triple (i, W, τ) , an actual persistence morphism*

$$\phi_{i,W,\tau} : M_{i,W,\tau} \longrightarrow N_{i,W,\tau}$$

is supplied. When source, target, kernel, and cokernel are terminal-tame on the declared scope, define

$$\mu_{i,W,\tau} := \text{tgdim}_W(\ker \phi_{i,W,\tau}), \quad \nu_{i,W,\tau} := \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}).$$

For a declared finite monitored set I_{mon} ,

$$\mu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \quad \nu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

The corresponding specialized status is exactly one of

$$\text{not_invoked}, \quad \text{undefined}, \quad \text{certified_zero}, \quad \text{obstructed}.$$

No numerical pair is imputed in the first two cases.

6 Toy Source Categories and Bridge Data

Definition 6.1 (Toy source category). *A toy source category is a declared small mathematical category or indexed diagram whose objects, morphisms, exact sequences, and colimits used by the bridge can be checked directly. The principal examples are:*

1. ToyVect_k : *finite-dimensional k -vector spaces;*
2. ToyMod_R : *finite-length left modules over an Artinian k -algebra R ;*
3. *finite barcode-profile diagrams realized as actual morphisms in $\text{Pers}_k^{\text{cons}}$;*
4. *finite direct sums, composable chains, and short exact comparison diagrams built from the preceding objects.*

Definition 6.2 (Toy bridge datum). *A toy bridge datum is a tuple*

$$\mathfrak{T} = (\mathcal{T}, X_{\bullet}, X_{\infty}, \mathcal{R}_{\text{Toy}}, I_{\text{mon}}, W, \tau, Q_{\bullet}, Q_{\infty}, \phi, \mathcal{E}, \mathcal{A}),$$

where:

1. $\mathcal{T}$ *is the toy source category;*
2. $X_{\bullet}$ *is a directed toy tower and X_{∞} is its declared apex;*
3. $\mathcal{R}_{\text{Toy}}$ *is the declared realization route;*
4. $(I_{\text{mon}}, W, \tau)$ *is the immutable monitored scope;*
5. $Q_n^{i,W,\tau}$ *and $Q_{\infty}^{i,W,\tau}$ are repaired Core-readable profiles;*
6. ϕ *is the actual tower comparison artifact when Type IV is invoked;*
7. $\mathcal{E}$ *is the evidence bundle;*
8. $\mathcal{A}$ *is the immutable artifact and provenance bundle.*

Definition 6.3 (Core-readable toy realization). *A realization route $\mathcal{R}_{\text{Toy}}$ is Core-readable on the declared scope when it supplies actual objects and morphisms in $\text{FiltCh}(k)$ or $\text{Pers}_k^{\text{cons}}$, repaired evaluations exist, constructibility is verified, and every invoked representative, eligibility, tower, or metric obligation is supplied as a separately typed record.*

Proposition 6.4 (Finite barcode readability). *Let Q be a finite barcode profile with declared endpoint convention, right-open window W , and threshold $\tau > 0$. Then $T_\tau^{\text{bar}}(W_W Q)$ is again a finite barcode profile. A tower of such profiles is Core-readable only after its transition data are realized as actual persistence morphisms or are explicitly restricted to object-level, non-Type-IV use.*

Proof. Window restriction clips finitely many intervals and discards empty intersections. Hard deletion removes a submultiset of the resulting finite multiset. Thus the output is finite. The final qualification is necessary because object-level barcodes alone do not manufacture the actual morphism required by Type IV diagnostics. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-003
Role / status	toy_bridge_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-008, CM-v20-009, CM-v20-022
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-003; conservatively inherited and re-bound to v20 sources.

7 Exact Tower Artifacts and Eventual Constancy

Definition 7.1 (Exact toy tower artifact). *For a repaired profile tower*

$$Q_0 \longrightarrow Q_1 \longrightarrow Q_2 \longrightarrow \cdots$$

with declared apex Q_∞ , an exact toy tower artifact is an actual morphism

$$\phi^{\text{stab}} : \text{colim}_n Q_n \longrightarrow Q_\infty$$

*constructed from a declared colimit and an artifact-backed apex agreement. Its runtime comparison mode is **exact**.*

Theorem 7.2 (Eventual-constancy toy bridge). *Let $Q_0 \rightarrow Q_1 \rightarrow \cdots$ be a repaired profile tower in a category admitting the declared colimit. Suppose:*

1. *there exists N such that every transition $Q_n \rightarrow Q_{n+1}$ is an isomorphism for $n \geq N$;*
2. *an isomorphism $a : Q_N \xrightarrow{\sim} Q_\infty$ is declared;*
3. *ϕ^{stab} is the colimit-to-apex morphism induced by the stabilized diagram and a ;*
4. *when Type IV is invoked, the source, target, and defect profiles are terminal-tame on the same monitored scope.*

Then ϕ^{stab} is an isomorphism. The Type IV status is *certified_zero* and

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

Proof. The colimit of an eventually constant diagram is canonically isomorphic to the stabilized value Q_N . Composition with a gives an isomorphism to Q_∞ . Kernel and cokernel vanish. Terminal-generic dimensions therefore vanish in every monitored degree. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-004
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-031
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-004; conservatively inherited and re-bound to v20 sources.

Corollary 7.3 (Finite barcode-profile stabilization). *Suppose a tower of finite barcode profiles is realized as an actual tower in $\text{Pers}_k^{\text{cons}}$, becomes isomorphic after a finite stage, admits the declared colimit, and satisfies apex agreement. Then the induced exact tower artifact is an isomorphism and its valid terminal-generic Type IV diagnostics vanish.*

Proof. Apply the eventual-constancy toy bridge to the realized persistence tower. The realization and actual-morphism hypotheses are essential; a sequence of multisets without morphisms is insufficient. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-005
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	TB-v20-003, TB-v20-004
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-005; conservatively inherited and re-bound to v20 sources.

Theorem 7.4 (Exact-conjugacy invariance of toy diagnostics). *Let $\phi : M \rightarrow N$ and $\phi' : M' \rightarrow N'$ be actual toy comparison artifacts. Suppose there are isomorphisms $\alpha : M \xrightarrow{\sim} M'$ and $\beta : N \xrightarrow{\sim} N'$ with*

$$\beta \circ \phi = \phi' \circ \alpha.$$

Then $\ker \phi \cong \ker \phi'$ and $\text{coker } \phi \cong \text{coker } \phi'$. On a common valid terminal-tame scope, all degree-wise and aggregated μ/ν diagnostics agree.

Proof. The source isomorphism restricts to kernels and the target isomorphism descends to cokernels. Terminal-generic dimension is invariant under persistence-module isomorphism. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-006
Role / status	toy_bridge_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-033
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-006; conservatively inherited and re-bound to v20 sources.

8 Finite-Dimensional Vector-Space Toy Towers

Theorem 8.1 (Monomorphism stabilization in finite dimension). *Let*

$$V_0 \xrightarrow{f_0} V_1 \xrightarrow{f_1} V_2 \rightarrow \dots$$

be a tower in ToyVect_k . If every f_n is injective and there is a finite bound D with $\dim_k V_n \leq D$ for all n , then the tower is eventually a tower of isomorphisms.

Proof. The integer sequence $\dim_k V_n$ is nondecreasing and bounded by D , hence eventually constant. An injective linear map between finite-dimensional vector spaces of equal dimension is an isomorphism. $\square$

Theorem 8.2 (Epimorphism stabilization in finite dimension). *Let $V_0 \rightarrow V_1 \rightarrow \dots$ be a tower in ToyVect_k whose transition maps are all surjective. Then it is eventually a tower of isomorphisms.*

Proof. The nonnegative integer sequence $\dim_k V_n$ is nonincreasing, hence eventually constant. A surjective map between finite-dimensional spaces of equal dimension is an isomorphism. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-007
Role / status	toy_source_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	local finite-dimensional proof
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T5
Repair ancestry	TB-v19-007; conservatively inherited and re-bound to v20 sources.

Corollary 8.3 (Vector-space stabilization bridge). *Let a vector-space toy tower satisfy either preceding stabilization theorem. Suppose $\mathcal{R}_{\text{Toy}}$ sends the eventual source isomorphisms to repaired profile isomorphisms, the repaired tower admits the declared colimit, and apex agreement and terminal-tameness records are supplied. Then its exact tower artifact has status **certified_zero**.*

Proof. The source theorem gives eventual isomorphism. The declared realization transfers this to eventual constancy of repaired profiles. Apply the eventual-constancy toy bridge. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-008
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	TB-v20-004, TB-v20-007
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-008; conservatively inherited and re-bound to v20 sources.

9 Finite-Length Module Toy Towers

Let R be an Artinian k -algebra and $\text{len}_R(-)$ the composition length.

Theorem 9.1 (Monomorphism stabilization in finite length). *Let*

$$M_0 \hookrightarrow M_1 \hookrightarrow M_2 \hookrightarrow \dots$$

be a tower of finite-length left R -modules. If $\text{len}_R(M_n) \leq L$ for a fixed finite L , then every transition is an isomorphism after some finite stage.

Proof. The length sequence is nondecreasing and bounded in $\mathbb{N}$, hence eventually constant. In a short exact sequence

$$0 \rightarrow M_n \rightarrow M_{n+1} \rightarrow C_n \rightarrow 0,$$

length additivity gives $\text{len}_R(C_n) = \text{len}_R(M_{n+1}) - \text{len}_R(M_n)$. At a constant stage this is zero, so $C_n = 0$ and the transition is an isomorphism. $\square$

Theorem 9.2 (Epimorphism stabilization in finite length). *Let $M_0 \twoheadrightarrow M_1 \twoheadrightarrow \dots$ be a tower of finite-length left R -modules. Then it is eventually a tower of isomorphisms.*

Proof. The nonnegative integer sequence $\text{len}_R(M_n)$ is nonincreasing and eventually constant. Length additivity for the kernel of a surjection then forces the kernel to vanish. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-009
Role / status	toy_source_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	local Jordan-Hoelder length proof
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T5
Repair ancestry	TB-v19-009; conservatively inherited and re-bound to v20 sources.

Corollary 9.3 (Finite-length module stabilization bridge). *Suppose a finite-length module tower satisfies either stabilization theorem, $\mathcal{R}_{\text{Toy}}$ sends eventual module isomorphisms to repaired profile isomorphisms, and the colimit, apex agreement, exact artifact, and terminal-tameness records exist. Then the monitored Type IV status is **certified_zero**.*

Proof. Apply the source stabilization theorem and then the eventual-constancy toy bridge. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-010
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	TB-v20-004, TB-v20-009
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-010; conservatively inherited and re-bound to v20 sources.

Remark 9.4 (Iwasawa boundary). *A finite-length module toy tower is not an Iwasawa tower. No passage from finite-length stabilization to an Iwasawa control theorem, information-preserving inverse/direct limit, characteristic ideal, or p -adic L -function statement is claimed.*

10 Eligible Degree-One Toy Edge

Theorem 10.1 (One-way eligible toy PH_1 -to- Ext^1 bridge). *Let a toy source object X admit a Core-readable filtered realization $F = \mathcal{R}_{\text{Toy}}(X)$. Suppose the complete eligible regime is verified: an admissible representative $C_{\tau,W}F$ exists, its monitored profiles agree with the repaired evaluations, its realization*

$$A := \mathcal{R}_{\text{Core}}(C_{\tau,W}F) \in D^{[-1,0]}(k\text{-mod})$$

is defined, and the degree-one edge-realization record identifies

$$H^{-1}(A) \cong H_1^{\tau,W}(F).$$

If

$$\text{Eval}_{1,W,\tau}(F) = 0,$$

then

$$\text{Ext}^1(A, k) = 0.$$

Proof. This is the v20 conditional one-way repaired PH_1 -to- Ext^1 bridge applied to the declared toy realization. The toy origin does not weaken any eligibility hypothesis. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-011
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-019, CM-v20-020
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-HIGHER-EDGE, FR-T5
Repair ancestry	TB-v19-011; conservatively inherited and re-bound to v20 sources.

Theorem 10.2 (Representative invariance of the eligible toy discharge). *Under the hypotheses of the preceding theorem, suppose A and A' are two eligible realized representatives connected by an artifact-backed isomorphism compatible with the degree-one edge records. Then*

$$\text{Ext}^1(A, k) = 0 \iff \text{Ext}^1(A', k) = 0.$$

Proof. Precomposition by the isomorphism induces an isomorphism of the corresponding Ext^1 groups. Compatibility of edge records ensures that both groups discharge the same repaired degree-one clause. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-012
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-032
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-HIGHER-EDGE, FR-T5
Repair ancestry	TB-v19-012; conservatively inherited and re-bound to v20 sources.

Guard-rail. No reverse implication and no global equivalence $\text{PH}_1 \cong \text{Ext}^1$ is proved here. The theorem is one-way and eligibility-dependent.

11 Terminal-Generic Defect Calculus

Theorem 11.1 (Terminal-generic index identity). *Let $\phi : M \rightarrow N$ be an actual morphism between terminal-tame persistence modules with finite-dimensional stabilized tails. Then*

$$\mu(\phi) - \nu(\phi) = \text{tgdim}(M) - \text{tgdim}(N).$$

If the tail dimensions agree, then $\mu(\phi) = \nu(\phi)$.

Proof. On a common stabilized tail, ϕ is a finite-dimensional linear map. Rank-nullity gives the displayed identity. The stabilized dimensions are the terminal-generic dimensions. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-013
Role / status	toy_bridge_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-034
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	TB-v19-013; conservatively inherited and re-bound to v20 sources.

Theorem 11.2 (Finite direct-sum additivity). *For terminal-tame actual morphisms $\phi_j : M_j \rightarrow N_j$, $1 \leq j \leq m$,*

$$\mu\left(\bigoplus_{j=1}^m \phi_j\right) = \sum_{j=1}^m \mu(\phi_j), \quad \nu\left(\bigoplus_{j=1}^m \phi_j\right) = \sum_{j=1}^m \nu(\phi_j).$$

Consequently, a finite direct sum is terminal-generically clean exactly when every summand is clean.

Proof. Kernels and cokernels commute with finite biproducts, and terminal-generic dimension is additive on finite direct sums. Nonnegativity gives the “exactly when” statement. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-014
Role / status	toy_bridge_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-035
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	TB-v19-014; conservatively inherited and re-bound to v20 sources.

Theorem 11.3 (Composition-controlled toy bridge). *Let*

$$M \xrightarrow{f} N \xrightarrow{g} P$$

be terminal-tame actual morphisms on a common terminal scope. Then

$$\mu(gf) \leq \mu(f) + \mu(g), \quad \nu(gf) \leq \nu(f) + \nu(g).$$

In particular, two clean comparisons have a clean composite.

Proof. On the stabilized tail, the kernel inequality follows from the map $\ker(gf) \rightarrow \ker g$, $x \mapsto f(x)$, whose kernel is $\ker f$. The cokernel inequality is the finite-dimensional rank inequality. Terminal-generic dimensions equal the tail dimensions. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-015
Role / status	toy_bridge_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-036
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	TB-v19-015; conservatively inherited and re-bound to v20 sources.

Remark 11.4 (No factorwise converse). *Cleanliness of gf does not imply cleanliness of f or g . Composition may kill or bypass a defect.*

Theorem 11.5 (Short-exact-diagram toy bridge). *Consider a commutative diagram in $\text{Pers}_k^{\text{cons}}$ with exact rows*

$$\begin{array}{ccccccccc} 0 & \rightarrow & M' & \rightarrow & M & \rightarrow & M'' & \rightarrow & 0 \\ & & \downarrow \phi' & & \downarrow \phi & & \downarrow \phi'' & & \\ 0 & \rightarrow & N' & \rightarrow & N & \rightarrow & N'' & \rightarrow & 0. \end{array}$$

Assume all six modules and the three vertical defect profiles are terminal-tame on one common scope. Then

$$\mu(\phi) \leq \mu(\phi') + \mu(\phi''), \quad \nu(\phi) \leq \nu(\phi') + \nu(\phi'').$$

Thus cleanliness of the two outer comparisons implies cleanliness of the middle comparison.

Proof. The snake lemma gives an exact sequence from the three kernels and cokernels. Passing to the common stabilized tail and taking dimensions yields the inequalities. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-016
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-037
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-T5
Repair ancestry	TB-v19-016; conservatively inherited and re-bound to v20 sources.

12 Single-Morphism Metric Bridge

Definition 12.1 (Finite-shift triviality). *For $\epsilon \geq 0$, a persistence module E is ϵ -trivial if its translation morphism $E \rightarrow E(\epsilon)$ is zero.*

Theorem 12.2 (Single-morphism interleaving bridge). *Let M, N be pointwise finite-dimensional one-parameter persistence modules and let $\delta \geq 0$. Then M and N are δ -interleaved if and only if there exists a morphism*

$$f : M \longrightarrow N(\delta)$$

whose kernel and cokernel are both 2δ -trivial.

Proof. This is the single-morphism characterization consumed by CM-v20-038, with its pointwise finite-dimensional and one-parameter hypotheses retained verbatim in scope. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-017
Role / status	toy_bridge_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-038
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-017; conservatively inherited and re-bound to v20 sources.

Theorem 12.3 (Finite-shift defect implies terminal-generic cleanliness). *Let $f : M \rightarrow N$ be an actual morphism whose kernel and cokernel are interval-decomposable, terminal-tame, and ϵ -trivial for a finite ϵ . Then*

$$\mu(f) = \nu(f) = 0.$$

Consequently, a single-morphism interleaving witness from the preceding theorem is terminal-generically clean when used as the declared Type IV comparison artifact on the same valid scope.

Proof. A terminal-cofinal interval summand has nonzero structure maps across every finite shift sufficiently far out, so it cannot occur in a finite-shift-trivial module. Hence neither defect profile has a terminal-cofinal summand. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-018
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-039
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	TB-v19-018; conservatively inherited and re-bound to v20 sources.

Theorem 12.4 (Cleanliness does not imply quantitative interleaving). *For every $\delta > 0$, terminal-generic cleanliness of an actual comparison morphism does not imply that its kernel and cokernel are 2δ -trivial and does not imply a δ -interleaving.*

Proof. Choose $L > 2\delta$, let $M = 0$, $N = k_{[0,L]}$, and take the zero morphism $f : M \rightarrow N(\delta)$. Its kernel is zero and its cokernel has no terminal-cofinal bar, so $\mu(f) = \nu(f) = 0$. But the cokernel is not 2δ -trivial and $d_I(M, N) = L/2 > \delta$. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-019
Role / status	toy_separation_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-040
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	TB-v19-019; conservatively inherited and re-bound to v20 sources.

13 Controlled Diagnostic Defect and Zero Defect

Definition 13.1 (Diagnostic dimension budget). *A diagnostic dimension budget is a pair of nonnegative integer bounds*

$$b_{i,W,\tau}^\mu, \quad b_{i,W,\tau}^\nu$$

satisfying

$$\mu_{i,W,\tau} \leq b_{i,W,\tau}^\mu, \quad \nu_{i,W,\tau} \leq b_{i,W,\tau}^\nu.$$

These are bounds on terminal-generic dimensions. They are not bottleneck distances and are not elements of the metric quantale unless an independent typed conversion theorem is supplied.

Proposition 13.2 (Bounded diagnostic defect is controlled, not clean). *Finite diagnostic dimension budgets imply controlled terminal-generic defect. They do not imply*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

unless every relevant zero conclusion is separately established.

Proof. A positive finite upper bound permits a positive finite diagnostic. Boundedness and vanishing are different predicates. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-020
Role / status	toy_controlled_defect_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-034, CM-v20-035, CM-v20-036, CM-v20-037
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	TB-v19-020; conservatively inherited and re-bound to v20 sources.

Proposition 13.3 (Zero diagnostic budget implies cleanliness). *If every invoked valid degree-wise diagnostic satisfies*

$$b_{i,W,\tau}^\mu = b_{i,W,\tau}^\nu = 0,$$

then the status is `certified_zero` and

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

Proof. The diagnostics are nonnegative integers bounded above by zero. They therefore vanish degree-wise and after finite aggregation. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-021
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-023
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-T5
Repair ancestry	TB-v19-021; conservatively inherited and re-bound to v20 sources.

Policy 13.4 (Rejection of the legacy mixed ledger). *The legacy expression*

$$\delta^\mu + \delta^\nu + \delta^{\text{real}} + \delta^{\text{apex}} < \text{gap}_\tau$$

is not a well-typed v20 statement when the first two terms are dimension bounds and the latter terms are metric discrepancies. TB-v20 replaces it by two independent records:

1. a diagnostic-dimension record controlling μ and ν ;
2. a metric ledger $\mathfrak{d}_{\text{met}} \in V$ with scalarization $\text{val}_B(\mathfrak{d}_{\text{met}})$ controlling an actual bottleneck discrepancy.

Neither record repairs a missing actual morphism, apex agreement, terminal-tameness certificate, realization, or artifact identity.

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-022
Role / status	metric_repair_policy / operational
Invocation / conformance	operational_rule / conformant
Source bindings	CM-v20-013, CM-v20-028, CR-G-010
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T5
Repair ancestry	TB-v19-022; conservatively inherited and re-bound to v20 sources.

14 Metric-Gap-Safe Toy Comparison

Theorem 14.1 (Sharp two-sided metric toy bridge). *Let B, C be finite barcodes, possibly with infinite bars, and let $\tau > 0$, $\varepsilon \geq 0$. Suppose*

$$d_B(B, C) \leq \varepsilon$$

and no finite bar of either barcode has length in the closed band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

Proof. A bottleneck matching of cost at most ε cannot match a retained finite bar on one side to a deleted finite bar on the other under the two-sided closed-band exclusion. Infinite bars match only infinite bars and are retained. Restricting the matching gives the asserted bound. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-023
Role / status	metric_repair_toy_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-012
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_metric_gap_certified
Allowed runtime modes	metric_gap_certified
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-023; conservatively inherited and re-bound to v20 sources.

Theorem 14.2 (Scalarized metric-ledger toy audit). *Let $(V, \oplus, 0_V, \preceq)$ be the declared metric quantale and*

$$\text{val}_B : V \rightarrow [0, +\infty]$$

a monotone subadditive scalarization. Let $\mathfrak{d}_{\text{met}}$ be the aggregate metric ledger element for a comparison of the windowed pre-threshold profiles B and C . Suppose the evidence bundle proves

$$d_B(B, C) \leq \varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}})$$

and

$$\varepsilon_B < \text{gap}_\tau(B, C) := \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\}.$$

Then the post-deletion comparison is metric-gap-certified and

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon_B.$$

This certifies the stated metric transport only. It does not establish Type IV cleanliness, theorem truth, or any missing non-scalar obligation.

Proof. The strict inequality gives the required two-sided avoidance of the closed threshold band. Apply the sharp metric toy bridge. The final sentence follows from the v20 separation of metric and non-scalar obligations. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-024
Role / status	conditional_metric_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-013, CM-v20-028
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_metric_gap_certified
Allowed runtime modes	metric_gap_certified
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-024; conservatively inherited and re-bound to v20 sources.

Proposition 14.3 (Exact comparison and metric robustness are distinct). *An exact equality or profile isomorphism may certify a specific exact comparison without any positive metric reserve. It does not by itself certify robustness under positive perturbation. Conversely, a metric-gap-certified comparison does not manufacture an exact isomorphism.*

Proof. The two runtime modes have different evidence and conclusions. A profile can be exactly compared with itself even when a bar lies exactly at threshold, where positive gap is zero. A finite bottleneck bound does not imply isomorphism. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-025
Role / status	toy_separation_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-014, CM-v20-015
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-025; conservatively inherited and re-bound to v20 sources.

15 Finite Detector Toy Categories and Stage Boundaries

For an interval-decomposable persistence module M and a finite stencil S of pairs (s, t) with $s < t$, define

$$D_S(M) := \bigoplus_{(s,t) \in S} \text{im}(M(s \leq t)).$$

The detector is typed by its source stage and by the exact geometric and endpoint convention used to construct S .

Theorem 15.1 (Detector-stage separation counterexample). *Let $\tau > 0$ and $M = k_{[0,\tau]}$ under an endpoint convention in which the interval has threshold length exactly τ . The pre-threshold profile is nonzero, while $T_\tau^{\text{bar}} M = 0$. Hence a detector nonzero on the pre-threshold stage does not imply a nonzero repaired profile, and a post-threshold zero theorem cannot be applied to a pre-threshold value without its stage-transport theorem.*

Proof. Hard deletion removes finite bars of length at most τ . The two stages are therefore different objects even though they arise in one evaluation chain. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-031
Role / status	toy_counterexample / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-041, CR-G-004
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 15.2 (Complete birth-stencil toy detector). *Let M be a finite direct sum of half-open intervals $k_{[b,d]}$ on a bounded window and fix $\tau > 0$. Suppose the complete endpoint list is part of the toy source datum. For every distinct birth b supporting a bar of length $> \tau$, choose*

$$0 < \eta_b < \min\{d - b - \tau : [b, d] \text{ is born at } b, d - b > \tau\},$$

and include the pair $(b, b + \tau + \eta_b)$ in S_{birth} . Then

$$D_{S_{\text{birth}}}(M) = 0 \iff T_{\tau}^{\text{bar}} M = 0.$$

The endpoint list and positive excesses are part of the completeness witness.

Proof. A short bar cannot contain a pair separated by more than τ . Every retained bar contains the selected pair at its birth, so it contributes a nonzero image. Conversely, any nonzero summand of the detector comes from a bar containing such a pair and therefore of length greater than τ . $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-032
Role / status	toy_detector_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-047, CM-v20-048
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 15.3 (Decoration-invariant germ toy detector). *For a finite barcode with arbitrary finite-endpoint decorations, replace evaluation at a decorated birth by the canonical right germ and choose the same positive-excess witnesses as above. The direct sum of the germ-to-later-time images is zero if and only if the repaired barcode is zero. The conclusion is invariant under changing only finite endpoint decorations while keeping the geometric intervals and the declared germ convention fixed.*

Proof. The right germ records the interval immediately after its geometric birth and removes ambiguity caused by whether the birth endpoint itself is included. Positive excess again distinguishes retained from deleted bars. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-033
Role / status	toy_detector_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-049
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 15.4 (Coverage-certified finite-stencil toy detector). *Let M be a finite interval-decomposable pre-threshold profile. Suppose a finite stencil S satisfies:*

1. *every pair $(s, t) \in S$ has $t - s > \tau$;*
2. *every bar of M having length $> \tau$ contains at least one pair $(s, t) \in S$ in its geometric interior, with endpoint decorations handled by the declared germ convention.*

Then

$$D_S(M) = 0 \iff T_\tau^{\text{bar}} M = 0.$$

Proof. Condition (1) prevents a deleted bar from contributing. Condition (2) forces every retained bar to contribute to at least one image summand. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-034
Role / status	conditional_toy_detector_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-050
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T3, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 15.5 (Uniform-excess family mesh certificate). *Let $W = [a, b]$ be bounded and let $\mathcal{F}$ be a family of finite interval profiles such that every retained window-relative bar has length at least $\tau + \gamma$ for one uniform $\gamma > 0$. Choose $0 < h < \gamma$ and the augmented mesh*

$$S_h = \{\min(a + jh, b) : 0 \leq j \leq \lceil (b - a)/h \rceil\}.$$

Use all admissible pairs $(s, s + \tau)$ with $s \in S_h$ and $s + \tau < b$, together with the declared right-edge/germ policy for window-cofinal bars. Then the resulting family detector is complete on $\mathcal{F}$. The augmented mesh contains exactly $\lceil (b - a)/h \rceil + 1$ points.

Proof. For a retained bar born at u , an augmented mesh point lies in $[u, u + h)$; since $h < \gamma$, the point shifted by τ remains strictly before the death. The count follows directly from the construction, including the adjoined right endpoint. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-035
Role / status	conditional_toy_detector_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-051, CM-v20-052, CM-v20-075
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T3, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 15.6 (Family-relative MinCert in a coordinate toy class). *Fix m designated disjoint retained intervals $I_1, \dots, I_m$ and let*

$$\mathcal{F}_m = \{0, k_{I_1}, \dots, k_{I_m}\}.$$

Restrict the information class to coordinate stencils, where one query can test only one designated interval. Then every complete zero-reflecting detector on $\mathcal{F}_m$ requires at least m queries, and the m coordinate queries are sufficient. Hence

$$\text{MinCert}(\mathcal{F}_m) = m$$

relative to this declared information class.

Proof. If a coordinate is not queried, the zero object and the corresponding singleton interval give identical observed data. Thus at least m queries are necessary. Querying all coordinates distinguishes zero from every nonzero member. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-036
Role / status	toy_mincert_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-053
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 15.7 (Right-unbounded finite-detector no-go toy family). *Let $W = [0, \infty)$ and let $\mathcal{U} = \{0\} \cup \{k_{[b, b+L)} : b \geq 0\}$ for fixed $L > \tau$. No detector using a fixed finite set of parameter pairs is complete on $\mathcal{U}$.*

Proof. A finite stencil has a largest parameter. Choose b larger than every parameter appearing in the stencil. The interval $k_{[b, b+L)}$ is retained but vanishes at every sampled source and target, so it is indistinguishable from zero. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-037
Role / status	toy_no_go_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-054
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 15.8 (Raw pointwise detector rank instability). *For every $\varepsilon > 0$ there are finite barcodes B, C with $d_B(B, C) < \varepsilon$ and a fixed pointwise stencil whose ranks differ. Thus raw detector rank is not bottleneck-stable.*

Proof. Choose $0 < \eta < \varepsilon$, $B = k_{[0,1]}$, and $C = k_{[\eta, 1+\eta]}$. Their bottleneck distance is η . At the fixed pair $(0, 1/2)$, the image rank is one for B and zero for C . $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-038
Role / status	toy_counterexample / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-046
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

16 Metric-Safe Zero Verdicts

Theorem 16.1 (Metric-safe detector zero-verdict bridge). *Let B, C be finite windowed pre-threshold profiles with $d_B(B, C) \leq \varepsilon$. Assume the two-sided closed threshold-band exclusion for both profiles, $2\varepsilon \leq \tau$, and complete zero-reflecting detectors for both repaired profiles. Then the two detector values are simultaneously zero. The operational certificate uses the strict reserve $2\varepsilon < \tau$.*

Proof. The threshold-gap theorem transports the bottleneck bound through hard deletion. Every nonzero repaired profile has distance at least $\tau/2$ from zero. Hence under $2\varepsilon \leq \tau$, one repaired profile is zero exactly when the other is. Detector completeness converts these zero predicates to detector values. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-039
Role / status	conditional_metric_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-044, CM-v20-045, CM-v20-056
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_metric_gap_certified
Allowed runtime modes	metric_gap_certified
Failure routes	FR-MISSING, FR-T3, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

17 Higher Ext Toy Bridges and Synthetic Boundary

Theorem 17.1 (Synthetic higher Ext identity and nonpromotion boundary). *Let V be a finite-dimensional vector space and put $A_V := V[n]$ in the cohomological derived category, so $H^{-n}(A_V) =$*

V. Then

$$\text{Ext}^n(A_V, k) \cong V^*.$$

This algebraic identity does not by itself realize a persistence profile, prove representative independence, or establish a natural PH_n -to- Ext^n bridge outside the deliberately defined synthetic category.

Proof. Over a field, the canonical higher edge identifies $\text{Ext}^n(A_V, k)$ with $\text{Hom}_k(H^{-n}(A_V), k)$. The final sentence lists independent obligations absent from the construction. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-040
Role / status	toy_separation_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-057, CM-v20-060, CR-G-007
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-HIGHER-EDGE, FR-NONVERDICT, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 17.2 (Synthetic natural zero-reflection on a finite detector category). *Let $\mathcal{T}_{\text{det}}$ be a deliberately small category of finite profiles equipped with one fixed complete functorial detector $D: \mathcal{T}_{\text{det}} \rightarrow \text{Vect}_k^{\text{fd}}$. Define $A_n(Q) := D(Q)[n]$. Then, naturally in this toy category,*

$$T_{\tau}^{\text{bar}}Q = 0 \iff D(Q) = 0 \iff \text{Ext}^n(A_n(Q), k) = 0.$$

The theorem is synthetic and family-relative; it is not a natural persistence realization theorem.

Proof. The first equivalence is detector completeness. The second is finite-dimensional dual zero-reflection applied to the preceding synthetic identity. Functoriality is inherited from the declared detector functor. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-041
Role / status	conditional_synthetic_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-057, CM-v20-077, CM-v20-085
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-HIGHER-EDGE, FR-NONVERDICT, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 17.3 (Complete detector higher Ext toy discharge). *Let a toy filtered realization be standard n -eligible, with an artifact-backed higher edge identifying $H^{-n}(A_n)$ with the declared extractor of $\text{Eval}_{n,W,\tau}(F)$. If a complete detector certifies $\text{Eval}_{n,W,\tau}(F) = 0$, then*

$$\text{Ext}^n(A_n, k) = 0.$$

A reverse implication is available only when the exact toy family also carries a proved natural zero-reflection theorem.

Proof. Detector completeness gives repaired-profile zero. The standard eligible higher-edge theorem gives Ext vanishing. The reverse direction requires zero-reflection and is not supplied by eligibility alone. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-042
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-062, CM-v20-085
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-HIGHER-EDGE, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 17.4 (Representative invariance of higher toy discharge). *If two standard n -eligible toy representatives are connected by an artifact-backed realization isomorphism compatible with their higher edge certificates, then their Ext^n groups are isomorphic and the higher vanishing verdict is representative-independent on that recorded scope.*

Proof. Contravariant derived Hom carries the realization isomorphism to an Ext isomorphism. Compatibility identifies the edge-discharge records. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-043
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-061
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-HIGHER-EDGE, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

18 Transient Defects and Full-Isomorphism Boundaries

Theorem 18.1 (Terminal-zero transient-defect counterexample). *Let $W = [0, U)$ and $0 < L < U$. For the actual morphism*

$$0 \longrightarrow k_{[0,L)},$$

the kernel is zero and the cokernel is the nonzero finite interior bar. Hence the terminal-generic diagnostics satisfy $(\mu, \nu) = (0, 0)$, but the morphism is not an isomorphism.

Proof. The finite bar does not reach the terminal end of W , so its terminal-generic dimension is zero. It remains a nonzero cokernel object. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-044
Role / status	toy_counterexample / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-063, CM-v20-067, CM-v20-079
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T4, FR-TRANSIENT, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 18.2 (Complete transient-defect detector characterization). *Let $\phi : M \rightarrow N$ be an actual morphism whose kernel and cokernel are finite interval-decomposable profiles. Suppose complete stage-typed detectors cover every retained interior bar of both defect profiles. Then both transient detector values are zero if and only if the retained interior parts of $\ker \phi$ and $\operatorname{coker} \phi$ vanish.*

Proof. Apply the coverage-certified detector theorem separately at the kernel-defect and cokernel-defect stages. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-045
Role / status	conditional_toy_detector_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-064, CM-v20-081
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-TRANSIENT, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 18.3 (Terminal plus transient zero gives toy isomorphism). *Under the preceding hypotheses, assume additionally that the kernel and cokernel records are the full τ -reduced defect profiles at the repaired stage. If the valid terminal diagnostics vanish and the complete transient detectors vanish on those profiles, then ϕ is an isomorphism on the declared repaired scope.*

Proof. Terminal zero removes cofinal defect summands; transient zero removes every retained interior defect summand. Because the recorded defect profiles are already τ -reduced, no unrecorded short summand remains in the repaired scope. Hence the repaired kernel and cokernel vanish, so ϕ is an isomorphism there. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-046
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-066
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-TRANSIENT, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 18.4 (Object-distance control does not control defect profiles). *Object-level distance bounds alone do not bound kernels or cokernels of comparison morphisms. Indeed, on one nonzero finite persistence module M , the two morphisms id_M and $0 : M \rightarrow M$ have identical source and target objects, but the first has zero defects while the second has kernel and cokernel isomorphic to M .*

Proof. The source-source and target-target object distances are zero. Defects depend on the actual morphism, not only on its endpoint objects. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-047
Role / status	toy_counterexample / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-080
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

19 Detector Atlases, Restart, and Finite-to-Infinite Reduction

Theorem 19.1 (Detector-safe atlas globalization). *Let a finite family of windows cover the declared global audit scope. Suppose every globally retained bar is covered by at least one local detector witness that survives the window-first clipping convention, all overlap identifications and threshold reconciliations are certified, and each local detector is complete at its declared stage. Then simultaneous local zero implies global repaired-profile zero.*

Proof. If a global retained bar existed, the global coverage hypothesis would assign it to a local complete detector, which would be nonzero. This contradicts simultaneous local zero. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-048
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-068, CM-v20-083
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 19.2 (Local-zero and Restart handoff are nonautomatic). *Let $M = k_{[0,2)}$, $\tau = 3/2$, and partition the scope into $W_1 = [0, 1)$ and $W_2 = [1, 2)$. Each clipped bar has length 1 and is deleted locally, while the unwind window has length 2 and is retained globally. Thus local detector zero on every window does not imply global zero without a detector-safe coverage or Restart handoff theorem.*

Proof. This is a direct calculation of window-relative lengths. The example also shows why crop-collapse commutation cannot be assumed. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-049
Role / status	toy_counterexample / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-069, CM-v20-082
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T3, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 19.3 (Exact finite-to-infinite detector reduction). *Let $Q_0 \rightarrow Q_1 \rightarrow \dots$ be an eventually constant repaired profile tower with exact apex agreement, and let N be a stabilization stage. If a complete finite detector certifies $Q_N = 0$, then*

$$\operatorname{colim}_n Q_n \cong Q_N \cong Q_\infty = 0.$$

Conversely, under the same exact identifications, apex zero implies stable-stage zero.

Proof. The exact eventually constant reduction identifies the colimit and apex with the stable stage. Transport zero through these isomorphisms. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-050
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-070
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	fixed_exact
Allowed runtime modes	exact
Failure routes	FR-MISSING, FR-T3, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-RETURN-MISSING, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

Theorem 19.4 (Finite prefixes do not determine the infinite reduction). *For every finite N , there are two towers agreeing through stage N whose colimits differ: the zero tower and the tower with $Q_n = 0$ for $n \leq N$ and $Q_n = k_{[0,\infty)}$ for $n > N$, with the first later map $0 \rightarrow k_{[0,\infty)}$ and identities thereafter. Hence no finite-prefix rule proves an infinite reduction without additional stabilization or convergence data.*

Proof. The two towers have identical finite prefixes. Their colimits are respectively zero and $k_{[0,\infty)}$. □

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-051
Role / status	toy_no_go_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-071
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-T3, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-RETURN-MISSING, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

20 Complete Toy Problem-Demonstration Composition

Theorem 20.1 (Complete toy problem-demonstration composition). *Suppose a toy problem record contains separately proved arrows*

$$\begin{aligned}
 \text{source condition} &\Rightarrow \text{Core-readable realization} \\
 &\Rightarrow \text{finite detector certificate} \\
 &\Rightarrow \text{repaired-profile zero} \\
 &\Rightarrow \text{eligible higher Ext zero.}
 \end{aligned}$$

together with an exact finite-to-infinite reduction when the target is infinite and a separately proved Return arrow to the declared toy target predicate. Then the composite source-to-target implication is valid. Omitting any invoked arrow leaves the corresponding conclusion undefined rather than partially proved.

Proof. Compose the displayed implications and the optional reduction and Return arrows. The undefined clause is the layered non-compensation rule. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-052
Role / status	conditional_toy_composition_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	CM-v20-072, CR-G-001, CR-G-006
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-HIGHER-EDGE, FR-T4, FR-DETECTOR-CERTIFICATION, FR-DETECTOR-NONZERO, FR-NONVERDICT, FR-TRANSIENT, FR-RETURN-MISSING, FR-T5
Repair ancestry	new v20 toy theorem/counterexample.

21 Combined Audit Readiness Without Compensation

Definition 21.1 (Toy bridge obligation bundle). *A v20 toy bridge obligation bundle contains independent atoms for:*

1. *source-category theorem validity and realization typing;*
2. *detector source stage, endpoint convention, soundness, completeness, family membership, coverage, and excess provenance when invoked;*
3. *comparison mode and exact or metric evidence;*
4. *representatives, eligibility, higher-edge naturality, and actual tower morphisms when invoked;*
5. *terminal tameness, terminal diagnostics, and separately typed transient-defect records;*
6. *finite-to-infinite reduction and Return arrows when invoked;*
7. *metric ledger, scalarization, and threshold-gap records in metric mode;*
8. *claim status, artifact identity, source binding, repair ancestry, replay, and manifest consistency.*

Theorem 21.2 (Layered non-compensating toy readiness). *A toy bridge record is audit-ready only when every required obligation atom passes and every excluded atom has justified **not_invoked** status. Detector completeness cannot compensate for missing eligibility; higher Ext zero cannot compensate for a missing Return arrow; terminal Type IV zero cannot compensate for an undetected transient defect; and a complete manifest cannot compensate for a false theorem hypothesis.*

Proof. The v20 status calculus is conjunctive over required dependency closure. Each missing or failed typed arrow retains its own failure route and is not canceled by success elsewhere. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-026
Role / status	operational_theorem / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	CM-v20-029, CM-v20-072, CR-G-010
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-T5
Repair ancestry	TB-v19-026; conservatively inherited and re-bound to v20 sources.

22 Toy Bridge Manifest and Local Register

Definition 22.1 (Canonical toy bridge manifest). *A TB-v20 manifest records at minimum: immutable run and schema identities; exact source artifacts; toy category and realization; detector stages and completeness sources; windows, thresholds, endpoint and cofinal policies; comparison modes; representatives and higher-edge records; actual tower artifacts; terminal and transient defect records; finite-to-infinite and Return records when invoked; obligation statuses and failure routes; local TB/CM/CR references; repair ancestry; replay data; and explicit external-domain non-claims.*

Theorem 22.2 (Manifest necessity and sidecar register discipline). *A TB theorem may be consumed by a later Bridge Program only if its manifest and local register are complete, mutually consistent, replayable at the declared scope, and source-bound to the exact theorem revision used. Manifest completeness is necessary for auditable reuse but is not sufficient for mathematical truth. The canonical 53-item UID list and complete status axes are carried by the JSON/CSV sidecars rather than duplicated in the mathematical body.*

Proof. The listed records are necessary to reconstruct the exact claim, hypotheses, stage, artifacts, and permitted use. Serialization cannot make a false or unsupported theorem true. Keeping the complete register in validated sidecars prevents body/register divergence while preserving human-readable scope and count here. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-027
Role / status	operational_theorem / operational
Invocation / conformance	operational_rule / conformant
Source bindings	CR-G-002, CR-G-014, CR-G-015, CR-G-016, CM-v20-073, CM-v20-086
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-T5
Repair ancestry	TB-v19-027; conservatively inherited and re-bound to v20 sources.

Definition 22.3 (Stable local TB UID rule). *Every registered item has a stable UID*

$$ak-hpdst:20.0.0:TB-v20:TB-v20-NNN.$$

The appendix version and future CR snapshot names are not embedded as mutable UID suffixes. This revision contains 53 consecutive local items, *TB-v20-001* through *TB-v20-053*.

Evidence status	Count
proved	24
conditional	23
operational	5
specification / explicit non-claim	1

23 Immutable TB-v19 to TB-v20 Migration

Policy 23.1 (Immutable migration). *The v19 TB artifact and its v18.3 ancestry are not overwritten. Items TB-v19-001–030 are conservatively re-bound as TB-v20-001–030, with stronger v20 metadata and unchanged lower-strength conclusions. The former TB-v19-031 closure is replaced by TB-v20-053. New v20 content occupies TB-v20-031–052.*

Predecessor range	v20 disposition	Principal change
TB-v19-001–010	inherited / re-bound	v20 source identities, stable UID namespace, and conservative-extension metadata
TB-v19-011–012	inherited / re-bound	higher-edge boundary added without strengthening degree-one conclusions
TB-v19-013–025	inherited / re-bound	terminal/transient and repaired-zero separation made explicit
TB-v19-026–027	strengthened operationally	detector, reduction, Return, replay, and sidecar obligations added
TB-v19-028–030	preserved at non-claim or conditional strength	no arithmetic or PDE promotion
TB-v19-031	superseded closure	replaced by TB-v20-053 after all new items
none	new TB-v20-031–052	detector certificates/no-go, higher Ext toys, transient defects, atlas/Restart, reduction, and problem-demonstration composition

Remark 23.2 (No silent predecessor mutation). *A statement change, scope change, or new theorem receives a new v20 UID. Repair ancestry records correspondence but never identifies the v19 and v20 artifacts as one immutable object.*

24 Arithmetic and PDE Readiness as Explicit Non-Claims

Proposition 24.1 (Iwasawa-style pattern only). *The implication*

$$\begin{array}{c} \text{eventual repaired-profile} \\ \text{stabilization} \end{array} \implies \text{exact tower artifact} \implies (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

is a valid toy pattern. It is not an Iwasawa theorem and does not establish information preservation along an Iwasawa tower.

Proof. The toy implication is TB-v20-004. An Iwasawa application additionally requires an arithmetic source category, purpose-faithful realization, finite-stage-to-limit transport, actual comparison artifact, and a proof that the realization preserves the arithmetic information of interest. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-028
Role / status	non_claim_pattern / spec
Invocation / conformance	specification_target / conformant
Source bindings	TB-v20-004, CR-G-001, CR-G-006
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry	TB-v19-028; conservatively inherited and re-bound to v20 sources.

Proposition 24.2 (Bounded-control caution only). *A uniformly bounded family of toy terminal-generic defects is controlled but not necessarily clean. Therefore a future arithmetic control theorem with bounded finite-layer error cannot be interpreted as information-loss-free without an additional vanishing or finite-shift-triviality theorem.*

Proof. Apply TB-v20-020 and compare with TB-v20-018. Bounded dimension and finite-shift triviality are different hypotheses. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-029
Role / status	non_claim_caution / proved
Invocation / conformance	theorem_evidence / conformant
Source bindings	TB-v20-018, TB-v20-020
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry	TB-v19-029; conservatively inherited and re-bound to v20 sources.

Proposition 24.3 (Toy dissipative instance). *A finite-dimensional toy dissipative tower whose repaired profiles are eventually constant, with declared apex agreement and exact tower artifact, has clean terminal-generic diagnostics.*

Proof. The dissipative label contributes no proof step. The conclusion is exactly TB-v20-004 applied to the repaired profile tower. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-030
Role / status	conditional_toy_bridge_theorem / conditional
Invocation / conformance	theorem_evidence / conformant_with_hypotheses
Source bindings	TB-v20-004
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	not_applicable
Allowed runtime modes	none
Failure routes	FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry	TB-v19-030; conservatively inherited and re-bound to v20 sources.

Remark 24.4 (No PDE theorem). *The preceding proposition is not an energy inequality, compactness theorem, weak-solution theorem, uniqueness theorem, regularity theorem, blow-up exclusion, or Navier–Stokes result.*

25 Explicit Exclusions

No statement in this appendix asserts any of the following:

- global exactness, functoriality, or non-expansiveness of hard positive-threshold deletion;
- unconditional crop–collapse commutation or silent detector-stage transport;
- detector zero from soundness alone, a family-independent MinCert value, or a uniform finite detector on an unbounded-birth family;
- pointwise detector-rank stability under bottleneck perturbation;
- a natural persistence-to-higher-Ext bridge from a shifted synthetic vector space alone;
- reverse higher-Ext transport without proved natural zero-reflection;
- an automatic tower comparison artifact, finite-to-infinite reduction, Restart handoff, or Return Theorem;
- a numerical diagnostic when Type IV is `undefined` or `not_invoked`;
- full isomorphism from terminal-generic cleanliness alone;
- control of kernel or cokernel profiles from object-distance bounds alone;
- conversion among diagnostic dimensions, bottleneck distances, actual morphisms, eligibility, completeness, or artifact identity without a typed theorem;
- identification of μ_{Iw} with μ_{Collapse} , information-loss-free arithmetic projection, a PDE regularity result, or any external-domain conclusion from a toy/Core pass;
- theorem status from Search, AI, Companion, Spec, registration, successful compilation, or manifest completeness alone.

26 Closure, Provenance, and Status

Theorem 26.1 (TB-v20 package closure and promotion boundary). *Within the declared toy regimes, Appendix TB-v20.0.0 provides, at the exact registered strengths:*

1. *inherited exact tower, finite-dimensional, finite-length, degree-one eligible, defect, interleaving, metric-gap, and audit theorems;*
2. *finite birth, germ, stencil, family-mesh, MinCert, and unbounded-birth no-go detector results;*
3. *metric-safe zero-verdict transport without pointwise-rank stability;*
4. *synthetic and eligible higher-Ext toy bridges with explicit naturality and nonpromotion boundaries;*
5. *terminal/transient separation, complete transient detection, and conditional full-isomorphism upgrade;*
6. *detector-safe atlas globalization, Restart counterexamples, exact finite-to-infinite reduction, and finite-prefix no-go;*
7. *a complete compositional toy model of the Problem-Bridge/Return architecture;*
8. *a 53-item co-normative local register, immutable repair ancestry, replay and artifact closure, and explicit external non-claims.*

No local TB item is formally promoted by the CR-v20 baseline. Promotion requires a later immutable CR receipt binding the final TB artifact and its exact local register; such promotion governs use but does not create proof.

Proof. The mathematical items are proved in the preceding sections from the displayed hypotheses or imported CM micro-theorems. The governance conclusion follows from the CR baseline rule that downstream promotion is separate from baseline registration. $\square$

TB UID	ak-hpdst:20.0.0:TB-v20:TB-v20-053
Role / status	operational_package_closure / operational
Invocation / conformance	operational_rule / conformant
Source bindings	TB-v20-001–052, CM-v20-086, CR-G-017
Source binding	direct immutable artifact binding; CR promotion pending
Comparison policy	runtime_resolved
Allowed runtime modes	exact, metric_gap_certified, not_compared
Failure routes	FR-MISSING, FR-T3, FR-T4, FR-NONVERDICT, FR-TRANSIENT, FR-RETURN-MISSING, FR-T5
Repair ancestry	TB-v19-031 package-closure ancestry; rebuilt for v20.
Mathematical conformance	conformant at proved or displayed conditional toy strength
Local register	53 consecutive UIDs; TeX/JSON/CSV identity required
External-domain status	explicit non-claim
CM dependency	direct immutable artifact binding
CR status	promotion pending; baseline does not promote TB-v20

References

- [1] U. Bauer and M. Lesnick, *Induced matchings and the algebraic stability of persistence barcodes*, Journal of Computational Geometry 6(2) (2015), 162–191.